

– SAFETY DATA SHEET

NN-PyroShield STC-I

Metal-Oxide Based Intumescent Fire-Retardant Coating

Prepared in the format of GHS / the Chemicals (MSIHC) Rules

Document No.	Revision	Issue Date	Product Code
NNPL/SDS/PROD-08A	1.0	2 July 2026	STC-I

01 IDENTIFICATION

Field	Detail
Product name	NN-PyroShield STC-I
Product type	Water-based intumescent fire-retardant coating
Recommended use	Passive fire protection coating for steel, timber and building elements
Supplier	NilayNarayan Polychem LLP, Plot D-167 (P), Kandra Ind. Area, Govindpur, Jharkhand 826010
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02 HAZARD IDENTIFICATION

The product is a water-based coating of low acute hazard. It is not classified as flammable. Hazards relate mainly to mild irritation from the wet paint and spray mist.

- **Irritant** — May cause mild skin and eye irritation on prolonged contact.
- **Spray mist** — Spray mist may irritate the respiratory tract; avoid inhaling mist.
- **Low fire risk** — Not flammable (water-based); dried film burns only under sustained external fire.

Precautionary: avoid contact with skin and eyes; wear gloves and eye protection; ensure ventilation during spraying; keep out of reach of children.

03 COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	CAS No.	Remark
Ammonium polyphosphate (APP)	68333-79-9	Acid source; flame retardant
Pentaerythritol	115-77-5	Carbon source
Melamine	108-78-1	Blowing agent
Titanium dioxide / metal-oxide char reinforcers	13463-67-7	Char strengthener; opacifier
Polymer binder + water	—	Film former / carrier

04 FIRST-AID MEASURES

- **Inhalation** — Move to fresh air. If mist has been inhaled and irritation persists, seek medical advice.
- **Skin** — Wash skin with soap and water before the paint dries; remove contaminated clothing.
- **Eyes** — Rinse cautiously with water for several minutes; remove contact lenses if easy. Seek medical attention if irritation persists.
- **Ingestion** — Rinse mouth and drink water; do not induce vomiting; obtain medical attention.

05 FIRE-FIGHTING MEASURES

- **Media** — Product is non-flammable (water-based). Use media appropriate to the surrounding fire.
- **Hazards** — On sustained fire the dried film chars and intumesces; decomposition may release oxides of carbon, nitrogen and phosphorus and ammonia.
- **Protection** — Wear self-contained breathing apparatus in enclosed-space fires.

06 ACCIDENTAL RELEASE MEASURES

- **Precautions** — Wear gloves and eye protection; prevent slip hazard from wet coating.
- **Containment** — Contain the spill; absorb with sand or inert absorbent; collect for disposal before it dries.
- **Environment** — Prevent entry into drains and watercourses; wash residues with water.

07 HANDLING & STORAGE

- **Handling** — Handle with gloves and eye protection; ensure ventilation during spraying; wash after handling.
- **Storage** — Store cool, dry and covered between +5 °C and +40 °C; keep containers closed; protect from frost.
- **General** — Keep away from foodstuffs and out of reach of children.

08 EXPOSURE CONTROLS / PERSONAL PROTECTION

Control	Recommendation
Spray mist / dusts	Observe local nuisance-dust / mist limits
Respiratory protection	Mist mask (P2) during spray application
Hand protection	Rubber / nitrile gloves
Eye protection	Safety goggles
Engineering controls	Ventilation / local exhaust during application

09 PHYSICAL & CHEMICAL PROPERTIES

Property	Value
Appearance	Off-white viscous paste / coating
Odour	Slight
Base	Water-based
Volume solids	≈ 65 – 72 %
VOC content	≤ 100 g/l
Flammability	Non-flammable (wet); dried film is fire-retardant
Solubility in water	Dispersible (wet); water-resistant once cured

10 STABILITY & REACTIVITY

- **Stability** — Stable under normal storage; protect from frost (freeze-thaw damages the emulsion).
- **Avoid** — Frost, prolonged heat and contamination.
- **Incompatibles** — Strong acids, strong oxidisers.
- **Decomposition** — On combustion: oxides of carbon, nitrogen and phosphorus, and ammonia.

11 TOXICOLOGICAL INFORMATION

- **Irritation** – Mild irritant to skin and eyes; spray mist may irritate the respiratory tract.
- **Systemic** – Low acute toxicity; no chronic hazard expected under normal use with good hygiene.

12 ECOLOGICAL INFORMATION

Wet product may be harmful to aquatic life due to its content; prevent release of uncured coating to drains, soil and watercourses. Cured film is largely inert.

13 DISPOSAL CONSIDERATIONS

Allow surplus coating to dry/cure where safe, then dispose of the solid as inert waste per local regulation. Liquid waste and washings are to be handled through authorised waste handlers; do not discharge to drains.

14 TRANSPORT INFORMATION

Not classified as dangerous goods for transport (non-flammable water-based coating). No UN number assigned under normal classification. Confirm against current transport regulations before shipment.

15 REGULATORY INFORMATION

Water-based intumescent coating for passive fire protection. Fire performance certified to the applicable standards (BS 476 Part 20/21, IS 12777). Handle in accordance with applicable chemical-safety regulations.

16 OTHER INFORMATION

This SDS is provided for safe-handling guidance. Classification and exposure limits must be verified against the regulations in force at the place of use. Issued by the NilayNarayan Polychem LLP NABL-accredited laboratory.

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